Data Preprocessing in Data Science

Transforming Raw Data into Meaningful Insights

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Content

- What is Data Preprocessing
- Why is Data Preprocessing Important?
- Steps in Data Preprocessing
  - Data Cleaning
  - Data Integration
  - Data Transformation
  - Data Reduction
  - Data Splitting
- Summary

What is Data Preprocessing

- Data preprocessing is the process of preparing raw data for analysis by cleaning and transforming it into a usable format.
- Goal is to improve the quality of the data.
- Helps in handling missing values, removing duplicates, and normalizing data.
- Ensures the accuracy and consistency of the dataset.

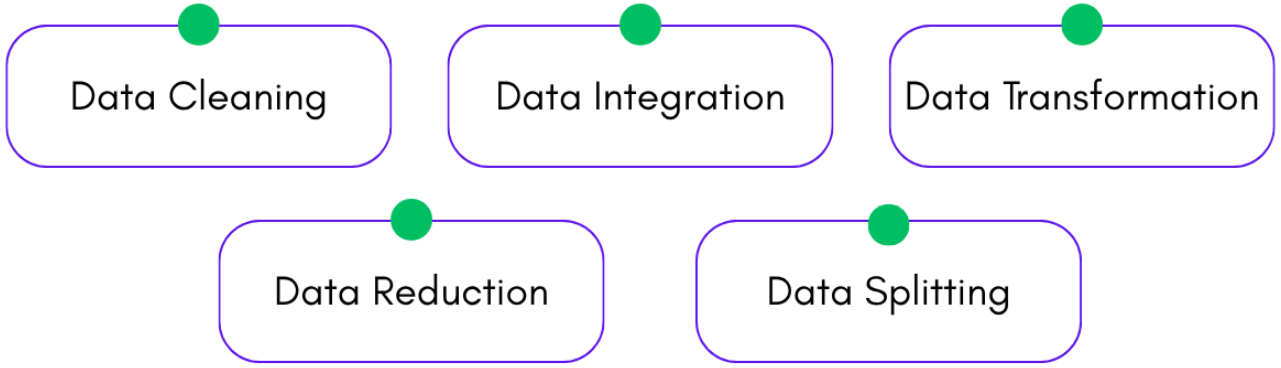

Figure 01: Steps in Preprocessing

Why is Data Preprocessing Important?

Raw data is often messy, containing errors, missing values, and inconsistencies. Data preprocessing helps identify and rectify these issues, resulting in a cleaner dataset that improves the quality of analysis.

- Raw data often contains missing values, duplicates, and inconsistencies.
- Machine learning models require numerical, clean, and normalized data.
- Improves accuracy, performance, and generalization of models.

MISSING VALUES

Working time [hours]	Sex	Annual salary
30	Male	$ 50 000
Null	Female	$ 125 000
25	Female	***Null***
50	Male	$ 160 000

Figure 02: Missing values Example

Steps in Data Preprocessing

- ❑ Data Cleaning
- ❑ Data Integration
- ❑ Data Transformation
- ❑ Data Reduction
- ❑ Data Splitting

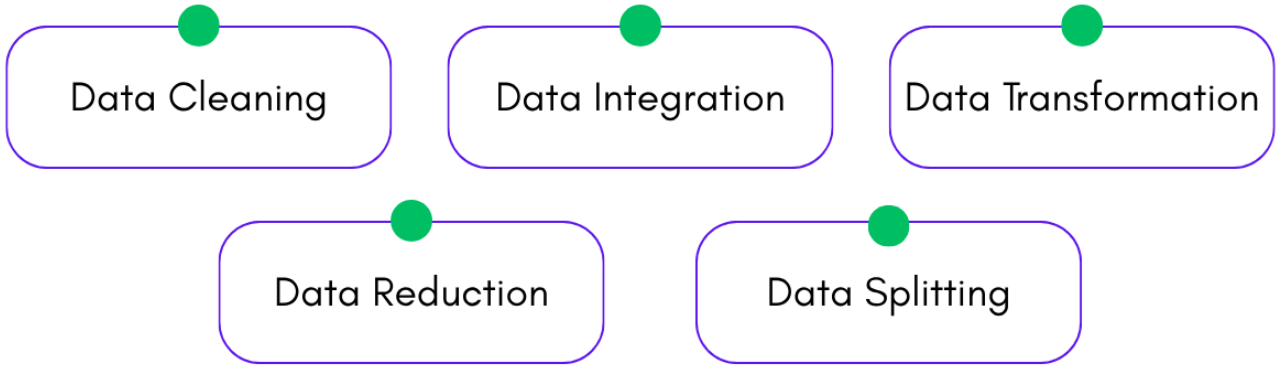

Figure 03: Steps in Preprocessing

Data Cleaning

- Handling missing values (drop or fill)
  - Replacing or removing data entries that have missing or null values to maintain dataset completeness.
- Removing duplicates
  - Eliminating repeated rows or records to ensure data accuracy and uniqueness.
- Fixing inconsistent formats
  - Standardizing data types, units, or formats (e.g., date, text case) to maintain uniformity across the dataset.

python

```
import pandas as pd

df = pd.read_csv('data.csv')

df.dropna(inplace=True)

df.drop_duplicates(inplace=True)
```

Data Transformation

Encoding categorical data

- ❑ Converting non-numerical (categorical) values into numerical form using techniques like LabelEncoder or OneHotEncoder for model compatibility.

Scaling numerical data

- ❑ Normalizing or standardizing numeric features using methods like StandardScaler or MinMaxScaler to ensure equal contribution to model training.

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python

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

df[['Age','Salary']] = scaler.fit_transform(df[['Age','Salary']])
```

Handling Missing Data

Remove missing rows

- ❑ Deletes all rows containing missing or null values from the dataset.

Fill with mean/median

- ❑ Replaces missing values with the column’s average or median value.

Forward fill/backward fill

- ❑ Fills missing values using the previous (forward) or next (backward) available data point.

Encoding Categorical Data

Label Encoding

- Converts categorical values into numeric labels by assigning each category a unique integer.

One-Hot Encoding

- Transforms categorical values into binary columns, creating a separate column for each category with 0s and 1s.

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python

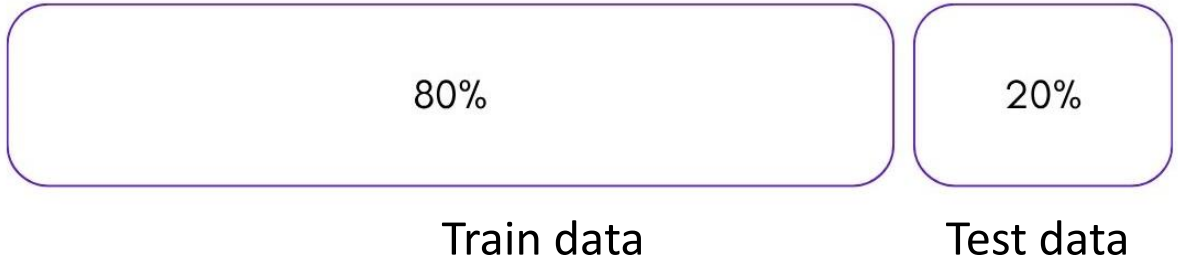
```
from sklearn.preprocessing import LabelEncoder

encoder = LabelEncoder()

df['Gender'] = encoder.fit_transform(df['Gender'])
```

Data Splitting

To separate data into training 80% and testing 20% sets for unbiased evaluation.




```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
```

Tools and Libraries

- ❑ **Python**
  - ❑for scripting
- ❑ **Pandas**
  - ❑for data manipulation
- ❑ **NumPy**
  - ❑for numerical operations
- ❑ **Scikit-learn**
  - ❑for preprocessing and ML tasks
- ❑ **Jupyter Notebook / Google Colab**
  - ❑for lab demonstrations

Summary

- ❑ Importance of Data Preprocessing
- ❑ Steps in preprocessing
- ❑ Practical Python examples
- ❑ Tools used for hands-on labs

Q&A: Any questions?